

Campaign B0: Exact Class-A Domain-Wall Calibration

Automated deterministic campaign report¹

¹*Repository experiment review*

(Dated: August 16, 2026)

This report records the fixed-rank CPU benchmark generated by campaign `implementation_smoke48`. It calibrates the analysis pipeline and is not evidence about the adaptive circuit. The transverse gate status is **FAIL**; the declared analysis width is $N_x = 6$.

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I. PROTOCOL AND PROVENANCE

Both the coupled spatial-mass interface and the hard-exterior analogue use the canonical `classA_U1FGTN._domain_wall_hamilton` source and exactly $N_x N_y$ occupied orbitals. The locked configuration and complete source hashes are in `manifest.json`. The width criterion is $m/(2\pi|v|/N_y) \leq 0.1$ at $N_y = 48$.

II. REPRESENTATIVE RESULTS

TABLE I. Accepted-width representative at $N_y = 48$.

construction	$ v $	m	width ratio	c_1	wall flows
coupled	0.98368	0.036	0.28	0.60251	-1, +1
hard_exterior	0.984517	0.00549	0.0426	1.0001	-1, +1

III. ENTROPY ACCOUNTING

The full-width strip has two entanglement boundaries and intersects both physical domain walls. Each three-column physical-wall contour window includes both entanglement cuts. Those window sums are diagnostic contributions to the full entropy, not isolated subsystem entropies. The saved products test equality of the contour sum and the entropy, equality of the two wall slopes, and the factor-of-two relation between a wall-window slope and the full-strip slope.

IV. ACCEPTANCE LEDGER

requirement	status	details
small-system formula and gauge preflight	PASS	completed before production stages
fixed-rank half filling	PASS	every geometry must contain exactly $N_x N_y$ occupied orbitals
Hermiticity, projector, and contour numerics	PASS	absolute tolerance 1e-10

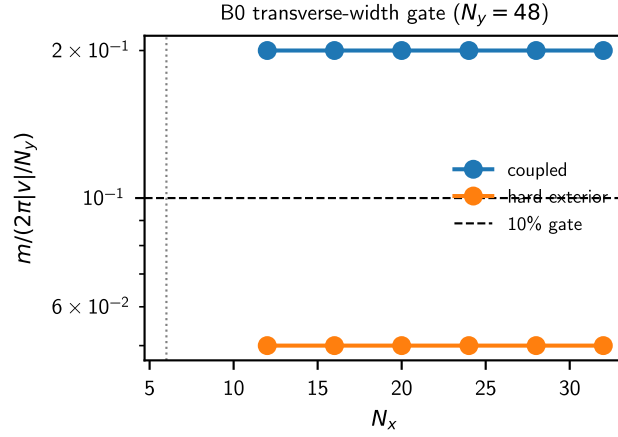


FIG. 1. Transverse-width gate. A failed gate is retained as a failed result rather than changing the threshold.

transverse-width hybridization gate FAIL
 coupled: opposite, compatible spectral PASS
 velocities
 coupled: finite transverse localization lengths FAIL
 coupled: Renyi $q=1,2,3$ full-strip $c=1$ FAIL
 coupled: single-physical-wall contour factor of two PASS
 coupled: wall squared-correlator exponent two FAIL
 coupled: physical response chirality PASS
 coupled: response linearity and charge conservation PASS
 coupled: opposite modular packet velocities PASS
 coupled: nonsingular overlap tracking PASS
 coupled: unit wall flow and zero net flow PASS
 hard exterior: opposite, compatible spectral velocities PASS
 hard exterior: finite transverse localization lengths FAIL
 hard exterior: Renyi $q=1,2,3$ full-strip $c=1$ PASS
 hard exterior: single-physical-wall contour factor of two PASS
 hard exterior: wall squared-correlator exponent two PASS
 hard exterior: physical response chirality PASS
 hard exterior: response linearity and charge conservation PASS
 hard exterior: opposite modular packet velocities PASS
 hard exterior: nonsingular overlap tracking PASS
 hard exterior: unit wall flow and zero net flow PASS
 33/65/129-point, both-sign, seam-relocation PASS
 twist validation

selected $N_x=6$; ratio threshold 0.1
 $v_0=-0.990652$, $v_1=0.990652$, relative mismatch= $2.24e-16$
 $xi_0=\text{nan}$, $xi_1=\text{nan}$
 $q=1$: $c=0.60251$, $q=2$: $c=0.53596$, $q=3$: $c=0.54232$
 relative errors= $[0.0012919593908469196, 0.0012919593904185955]$; each window includes both entanglement cuts
 $betas=[2.393660509217847, 2.3936605090610428]$; all four candidate models remain in correlator_fits.csv
 response velocities= $[-0.36468406133616255, 0.17330407060844988]$; spectral velocities= $[-0.990652363292239, 0.9906523632922388]$
 charge drift= $2.74e-16$; epsilon relative error= $5e-07$
 modular velocities= $[-0.028489122097027773, 0.007101623085203323]$
 minimum singular value= 0.99956
 signed flows= $[-1, 1]$
 $v_0=-0.990652$, $v_1=0.990652$, relative mismatch=0
 $xi_0=\text{nan}$, $xi_1=\text{nan}$
 $q=1$: $c=1.0001$, $q=2$: $c=0.99592$, $q=3$: $c=0.99375$
 relative errors= $[7.642432497467766e-05, 7.642432449883607e-05]$; each window includes both entanglement cuts
 $betas=[2.0742276445890773, 2.074227644533882]$; all four candidate models remain in correlator_fits.csv
 response velocities= $[-0.38144918141768897, 0.193895169143781]$; spectral velocities= $[-0.9906523632922394, 0.9906523632922394]$
 charge drift= $4.25e-16$; epsilon relative error= $5e-07$
 modular velocities= $[0.030084364672927803, 0.05346506281391794]$
 minimum singular value= 0.999999
 signed flows= $[-1, 1]$
 all validation rows require unit absolute wall flow and seam/uniform spectral agreement

The machine-readable ledger retains the same decisions and their thresholds. Any failed item remains explicit rather than changing the estimator or gate.